Choosing the Right Control Structure

Instructor Answer Key

Each option is the best answer at least once: for-each 2, while 2, do-while 2, for 2, if 1, none 1.

Quick reference

#	Scenario	Answer	Why
1	Shopping cart total	**A. for-each**	Visits every item in the list; no index is needed.
2	Game menu	**C. do-while**	The menu must appear at least once before the Quit check.
3	Free shipping	**E. if statement**	One decision, made once; nothing repeats.
4	Reverse order	**D. for**	Needs an index counting down from `length - 1`; for-each only goes forward.
5	Reading a file	**B. while**	Check before reading; an empty file means the body runs zero times.
6	Swap two values	**F. None of the above**	Three assignments with a temp variable; no decision or repetition.
7	Email blast	**A. for-each**	Every element, in order, no index.
8	Roll until six	**C. do-while**	You must roll before you can test for a 6.
9	Doubling savings	**B. while**	The number of years is unknown; loop while balance < $2,000.
10	Rocket countdown	**D. for**	A fixed count from 10 to 1 with a counter.

Scenarios with answers

1. Shopping cart total. An online store keeps the items in a customer's cart in an `ArrayList<Item>`. Add up the price of every item to get the cart total.

Answer: A. for-each — Visits every item in the list; no index is needed.

2. Game menu. A text game shows a menu (Play, Settings, Quit), reads the player's choice, and acts on it. The menu keeps reappearing until the player picks Quit. The menu must be shown at least once.

Answer: C. do-while — The menu must appear at least once before the Quit check.

3. Free shipping. After checkout, the program prints "You qualify for free shipping!" if the order total is $50 or more.

Answer: E. if statement — One decision, made once; nothing repeats.

4. Reverse order. Print the elements of an `int[] scores` array from the last element to the first.

Answer: D. for — Needs an index counting down from `length - 1`; for-each only goes forward.

5. Reading a file. A program reads lines from a text file and prints each one until there are no lines left. The file might be empty.

Answer: B. while — Check before reading; an empty file means the body runs zero times.

6. Swap two values. Two variables, a and b, hold numbers. Exchange their values so a holds what was in b and b holds what was in a.

Answer: F. None of the above — Three assignments with a temp variable; no decision or repetition.

7. Email blast. A club stores its members' email addresses in a list. Print every address so the secretary can copy them into an email.

Answer: A. for-each — Every element, in order, no index.

8. Roll until six. Simulate rolling a die and print each roll. Keep rolling until a 6 comes up. You can't know the result until you've rolled at least once.

Answer: C. do-while — You must roll before you can test for a 6.

9. Doubling savings. A savings account starts at $1,000 and earns 5% interest per year. Count how many years it takes for the balance to reach $2,000.

Answer: B. while — The number of years is unknown; loop while balance < $2,000.

10. Rocket countdown. Print a countdown from 10 down to 1, then print "Liftoff!"

Answer: D. for — A fixed count from 10 to 1 with a counter.

Discussion prompts

• Scenario 1 could be written with a for loop. Why is for-each still the better choice?

• Scenario 8 can be rewritten as a while loop. What extra code would that need?

• Scenario 3 uses an if statement but no loop. What changes if the program checks every order placed today?